

# **Project 1**

**DATA 422: Advanced Data Science Methods and Ethics**

**Due (full package):** Friday, October 2

**Weight:** 20% of the course (stepping stones count toward this project, not toward labs)

Wildfire insurance AI lab: a **risk model** (Model 1) and later an **evidence-to-text** model (Model 2). Weeks 1–2 are only Model 1.

# Target (do not loosen this language)

We estimate

$$\mathbb{P}(\text{serious\_damage} = 1 \mid \text{fire contact context}, s, x, \mathcal{F}_{\text{event}})$$

using CAL FIRE DINS structure, exposure, and hazard-context features (see [data card](#)).

**serious\_damage** is 1 for Major or Destroyed, 0 for No Damage / Affected / Minor. Inaccessible rows are out of the supervised table.

This is **not** an annual household loss probability, **not** a premium, and **not** a coverage decision. A separate rule (threshold / abstain / specialist review) will turn scores into an action. Model 2 may describe that action; it may not invent coverage.

# Stepping stones

| Stone                   | Due                   | What you turn in                                          |
|-------------------------|-----------------------|-----------------------------------------------------------|
| <a href="#">Stone 1</a> | Wed Sep 9, 11:59 p.m. | EDA + feature deep dive + first model on <b>selection</b> |

# Model 1 workflow

Use the modeling checklist from lecture: problem  $\rightarrow$  data  $\rightarrow$  model  $\rightarrow$  loss  $\rightarrow$  fit  $\rightarrow$  evaluate on **selection**. PyTorch is fine but not required for stone 1.

# Data

Download [properties\\_modeling\\_v4.csv](#) (also linked from [stepping stone 1](#)). The `split` column is pre-set:

| Split                   | Role                                                 |
|-------------------------|------------------------------------------------------|
| <code>train</code>      | Fit models; compute encoding / imputation statistics |
| <code>selection</code>  | Compare models and metrics                           |
| <code>assessment</code> | Final evaluation only (stone 2)                      |

Column definitions and primary source links: [Project 1 data card](#).

The table is built from public [CAL FIRE DINS](#) with joined hazard and fire-perimeter layers (see data card).

# Labs vs this project

Tuesday labs are ungraded **Colab** notebooks (practice for the Friday **paper** quiz). They are not stepping stones. The stones on this page are the project checkpoints.